Exploratory Data Analysis (EDA) Report

1. Dataset Loading & Initial Inspection

The dataset was loaded using the Pandas library. To understand its structure, the first few (df.head()) and last few (df.tail()) records were examined.

Observation:

- The dataset consists of **10,000 rows and 6 columns**.
- Each row represents a student's study-related attributes and their performance outcome.
- Data appears structured and consistent across records.

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2. Feature Type Identification (Manual Inspection)

Based on column names and values, features are categorized as follows:

Numerical Features

- **Hours Studied** – Number of hours spent studying.
- **Previous Scores** – Previous academic score.
- **Sleep Hours** – Average daily sleep duration.
- **Sample Question Papers Practiced** – Number of practice papers solved.
- **Performance Index** – Continuous performance score.

Categorical Feature

- **Extracurricular Activities** – Indicates participation (Yes / No).

Binary Feature

- **Extracurricular Activities** (Binary categorical: Yes/No)

Ordinal Features

- None explicitly present. (All numeric values represent quantities, not rankings.)

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3. Data Type & Statistical Summary Analysis

Using df.info()

- All columns have **non-null values (100%)**.
- Data types:
    - int64: 4 columns
    - float64: 1 column
    - object: 1 column
- Memory usage is low (~469 KB), making it computationally efficient.

Using df.describe()

Key statistical observations:

- **Hours Studied** ranges from **1 to 9**, with a mean of ~5.
- **Previous Scores** range from **40 to 99**, indicating moderate to high academic history.
- **Sleep Hours** average around **6.5 hours**, suggesting realistic student behavior.
- **Performance Index** ranges from **10 to 100**, suitable as a prediction target.

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4. Categorical Feature Distribution

For **Extracurricular Activities**:

- Two unique values: **Yes** and **No**
- Distribution is fairly balanced:
    - No: ~50.5%
    - Yes: ~49.5%

Observation:

- No major class imbalance exists in the categorical feature.

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5. Target Variable & Input Features

Target Variable

- **Performance Index**
    - Continuous numerical variable

    - Suitable for **regression-based machine learning models**

Input Features

- Hours Studied
- Previous Scores
- Extracurricular Activities
- Sleep Hours
- Sample Question Papers Practiced

These features are logically related to academic performance, making them strong predictors.

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6. Dataset Size & ML Suitability

- **10,000 records** is a solid dataset size.
- Adequate for:
    - Linear Regression
    - Decision Tree
    - Random Forest
    - Gradient Boosting models
- Low risk of overfitting with proper validation.

Conclusion: The dataset is **well-suited for machine learning**, especially supervised regression tasks.

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7. Data Quality Observations

Positive Aspects

✅ No missing values ✅ No duplicate records observed ✅ Balanced categorical feature ✅ Clean and well-structured dataset

Potential Considerations

⚠ Categorical column requires encoding before ML ⚠ Feature scaling may improve performance for distance-based models

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Final Conclusion

This dataset is **high-quality, clean, and ML-ready**. Minimal preprocessing is required, making it ideal for educational purposes, EDA practice, and regression model development.